Ordinary Differential Equations: Abridged Notes

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These abridged notes were reconstructed afterward because I did not take direct notes during class. They are not a record of the actual lecture.

Contents

1 Topic Summary **2**

2 From Complex Roots to Real Solutions **2**

2.1 Example: Damped Oscillation 2

2.2 Example: Growing Oscillation with Initial Data 2

1 Topic Summary

Complex roots of a real constant-coefficient characteristic equation still give real solutions. A conjugate pair $r = \alpha \pm i\beta$ produces $e^{\alpha t}\cos\beta t$ and $e^{\alpha t}\sin\beta t$. The real part α controls the amplitude: negative gives decay, positive gives growth, and zero gives constant-amplitude oscillation. Initial conditions fix the two constants.

2 From Complex Roots to Real Solutions

For $ay'' + by' + cy = 0$, solve $ar^2 + br + c = 0$. If the roots are $\alpha \pm i\beta$, with $\beta > 0$, Euler's formula gives the real general solution

$$\boxed{y(t) = e^{\alpha t}\big(C_1\cos\beta t + C_2\sin\beta t\big).}$$

The oscillation period is $2\pi/\beta$. The factor $e^{\alpha t}$ is an amplitude envelope, so the sign of α determines long-term growth or decay.

2.1 Example: Damped Oscillation

Adapted from Boyce et al., 11th ed., §3.3, Problem 8. Solve $y'' + 6y' + 13y = 0$.

Solution.

The characteristic equation has roots

$$r = \frac{-6 \pm \sqrt{36 - 52}}{2} = -3 \pm 2i.$$

Therefore

$$\boxed{y(t) = e^{-3t}\big(C_1\cos 2t + C_2\sin 2t\big).}$$

The solution oscillates with period π, while its amplitude shrinks like e^{-3t}.

2.2 Example: Growing Oscillation with Initial Data

Adapted from Boyce et al., 11th ed., §3.3, Problem 13. Solve

$$y'' - 2y' + 5y = 0, \qquad y(\pi/2) = 0, \qquad y'(\pi/2) = 2.$$

Solution.

The roots are $1 \pm 2i$. Center the solution at $t_0 = \pi/2$ by writing $s = t - \pi/2$:

$$y = e^{s}(A\cos 2s + B\sin 2s).$$

At $s = 0$, $y = A = 0$ and $y' = A + 2B = 2$, hence $B = 1$. Thus

$$\boxed{y(t) = e^{t-\pi/2}\sin\big(2(t - \pi/2)\big).}$$

It crosses zero periodically, but its oscillation envelope grows as t increases because the characteristic roots have positive real part.